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SPECIALIZATION IN CYBERSECURITY DEPARTMENT OF ENGINEERING
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Security is not Carbon-Free: Evaluating the Environmental Impact of WiFi Intrusion Detection Systems

A simulation-based experimental study

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Summary

Cybersecurity mechanisms are widely deployed to protect wireless networks, yet their environmental impact remains largely unexplored. As WiFi infrastructures continue to scale globally, the cumulative energy cost of security mechanisms may contribute non-negligibly to carbon emissions. This paper presents a carbon-aware evaluation of intrusion detection strategies in WiFi networks, aiming to quantify the environmental cost of security alongside its operational benefits. An ns-3-based simulation framework is developed to model a WiFi network comprising a central access point and multiple communicating nodes. The network is evaluated under baseline and abnormal traffic conditions, followed by the deployment of three intrusion detection strategies: static threshold-based, adaptive traffic-aware, and hybrid detection. The analysis considers energy consumption, network performance, and detection behavior, and translates energy usage into carbon emissions using region-specific carbon intensity factors. The results show that intrusion detection mechanisms introduce measurable energy overhead, leading to increased carbon emissions despite improving network robustness. Furthermore, the carbon footprint of identical detection strategies varies significantly across geographical regions due to differences in energy production profiles. Among the evaluated approaches, adaptive and hybrid strategies demonstrate improved operational efficiency but at varying environmental costs, highlighting inherent trade-offs between security effectiveness and sustainability. This study introduces carbon-aware evaluation as a complementary dimension in the design and assessment of intrusion detection systems. The findings emphasize that cybersecurity is not environmentally neutral and underline the need for sustainability-driven design principles in next-generation wireless security solutions.

Preface

First and foremost, we would like to express our sincere gratitude to our supervisor, Rashid Ali at University West, who contributed the original idea behind this work. Thank you for your invaluable input, guidance, and inspiration, which not only helped us complete this work, but also made the entire process both educational and enjoyable. Furthermore, we would also like to thank our course coordinators: Abdulghafour, Fatiha, Mats and Robert, and last but not least our examiner Vira. Thank you for all the knowledge you have shared and for making our time as students on the Master's program in Cybersecurity at University West, both valuable and memorable.

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Idrisa, thank you for a great collaboration and for always maintaining a positive attitude, even during challenging times. Working with you has been both enjoyable and educational, and I hope we meet again in the future. To my boyfriend, Max, and to my sister, Cornelia: thank you. / Alice.

Alice, / Idrisa

Affirmation

This master degree report, "Security is not Carbon-Free: Evaluating the Environmental Impact of WiFi Intrusion Detection Systems", was written as part of the master degree work needed to obtain a Master of Science with specialization in Cybersecurity degree at University West. All material in this report, that is not my own, is clearly identified and used in an appropriate and correct way. The main part of the work included in this degree project has not previously been published or used for obtaining another degree.

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1 | Introduction

1.1 Motivation

Wireless local area networks (WLANs) based on the IEEE 802.11 standard enable wireless communication between devices using radio-based transmission. In infrastructure mode, access points (APs) coordinate communication between wireless stations and act as intermediaries for data transmission [1, 2]. As the number of connected devices and the volume of wireless traffic continue to increase, Wi-Fi networks have evolved into critical infrastructure supporting both everyday communication and essential services. To ensure a secure operation, Wi-Fi networks rely on continuous traffic monitoring and intrusion detection systems (IDS), which aim to preserve the core security properties of availability, confidentiality, and integrity [3]. Monitoring techniques such as packet capture, flow-based analysis and anomaly detection are widely used to observe network behavior and identify potential threats [4, 5].

Wireless intrusion detection systems (WIDS) detect wireless attacks by analyzing network traffic and protocol behavior [6, 7]. Monitoring and detection systems use extra computing power and network resources. Activities like checking data packets, extracting information, and making real-time decisions require processing capacity and create more network traffic. Research shows that IDS, especially those using machine learning, can greatly increase CPU usage, system workload, and energy consumption in real-time settings [8].

Intrusion detection involves analyzing network traffic and system events, which requires processing large amounts of data. This makes detection increasingly complex as network activity grows [9, 10]. At the same time, the environmental impact of information and communication technology (ICT) is becoming more important. The ICT sector uses a significant amount of the world's electricity and produces carbon emissions [11]. Since energy use is directly linked to carbon emissions, systems that require a lot of computing power such as IDS can have a significant environmental impact, especially when they run continuously [12]. Signature-based methods detect known attack patterns, while anomaly-based methods identify deviations from normal behavior [4]. Hybrid approaches combine these methods to improve detection performance [13].

In practice, IDS are primarily evaluated using detection performance metrics such as accuracy, precision, recall, false positive rate, and detection delay [8, 14]. Additional system-level metrics, such as throughput and latency, are sometimes considered [7, 15]. Recent research has introduced sustainability-oriented perspectives, where energy consumption and carbon emissions are included in system evaluation [7, 16, 17]. Carbon-aware computing further links energy usage to carbon intensity, enabling evaluation of environmental impact [18].

Despite these developments, sustainability is not consistently integrated into intrusion detection evaluation. Most IDS studies still focus primarily on detection performance, while energy consumption and carbon emissions are often not systematically analyzed [9, 17]. Existing research on energy-aware IDS is largely concentrated in data centers, IoT systems, and edge environments, and often focuses on model-level optimization rather than system-level evaluation [19, 20, 21]. In contrast, Wi-Fi environments, and particularly access-point-level monitoring, remain less explored in terms of combined evaluation of detection performance, resource usage, and environmental impact.

Furthermore, the impact of monitoring placement on performance, energy consumption, and carbon emissions is not well understood in Wi-Fi networks. Although access-points (APs) play a central role in coordinating communication and observing traffic [1, 2], limited work has analyzed how IDS deployment at the AP level affects both system performance and sustainability.

The problem addressed in this thesis is that intrusion detection in Wi-Fi networks is primarily evaluated based on detection performance, while the associated energy consumption and carbon emissions are not systematically considered. This limits the ability to assess trade-offs between detection effectiveness, computational cost, and environmental impact. In particular, there is insufficient understanding of how access-point-level monitoring influences these trade-offs in practical Wi-Fi environments. To address this problem, this thesis investigates sustainable cybersecurity mechanisms for Wi-Fi networks, with a focus on access-point-level traffic monitoring and intrusion detection. A simulation-based experimental framework is developed in ns-3 to evaluate IDS under controlled conditions. The study measures detection performance, network performance, energy consumption, and estimated carbon emissions, enabling analysis of trade-offs between security effectiveness and sustainability.

1.2 Aim

The overall aim of this thesis is to design, measure, and evaluate sustainable cybersecurity mechanisms for Wi-Fi networks, with a specific focus on access-point-level traffic monitoring and intrusion detection.

The specific aims are:

- To measure the energy consumption and estimated carbon footprint of Wi-Fi traffic monitoring and intrusion detection at the access point level.
- To analyze how IDS design and monitoring configuration affect detection performance, resource use, and sustainability.
- To investigate whether adaptive or selective monitoring can reduce environmental impact while maintaining acceptable security performance.

1.3 Research questions

1. What is the energy consumption and estimated carbon footprint of common Wi-Fi security monitoring tasks?

2. How do IDS design and monitoring strategies impact detection performance and energy usage?
3. Can carbon-aware evaluation or selective monitoring reduce emissions without unacceptable security risk?

1.4 Contributions

The main contributions of this thesis are:

- A simulation-based evaluation framework in ns-3 for analyzing Wi-Fi intrusion detection at the access point level, including energy measurement and carbon emission estimation.
- A systematic comparison of intrusion detection approaches under controlled Wi-Fi traffic conditions, enabling analysis of performance–energy trade-offs.
- A sustainability-oriented evaluation methodology that integrates traditional detection metrics with system-level metrics such as energy consumption and estimated carbon footprint.
- A quantitative analysis of trade-offs between detection performance, computational cost, and environmental impact in Wi-Fi security monitoring.
- Empirical insights into resource-efficient and carbon-aware monitoring strategies that maintain acceptable levels of detection performance.

2 | Background

2.1 Wi-Fi Networks and Access Points

Wi-Fi networks are based on the IEEE 802.11 standard, which defines the medium access control (MAC) and physical layer (PHY) specifications for wireless local area networks [2]. In infrastructure mode, communication between wireless devices is coordinated through an AP, which acts as an intermediary and connects devices to the network through a distribution system[2]

An AP together with its connected devices forms a Basic Service Set (BSS), which is the basic unit of an IEEE 802.11 network. Multiple BSSs can be connected to form an Extended Service Set (ESS), which allows communication over a larger area [2]. In infrastructure networks, communication between devices is coordinated through the AP[1, 2]. This means that the AP plays a central role in handling network traffic. Because of this, the AP can observe the traffic exchanged between devices, which makes it relevant for traffic monitoring and security analysis.

Modern Wi-Fi standards are designed to support a large number of devices and high traffic demand. For example, IEEE 802.11ax (Wi-Fi 6) introduces mechanisms that improve resource efficiency and performance in dense network environments[22]

2.2 Traffic Monitoring in Wi-Fi

Communication in IEEE 802.11 networks is based on frame exchanges at the MAC layer. Communication in IEEE 802.11 networks is based on the exchange of frames between wireless devices and APs at the MAC layer [1, 12]. These frames can be observed and used to analyze network activity.

Packet-level monitoring captures individual packets and provides detailed information about communication patterns, protocol behavior, and traffic characteristics [12]. However, capturing and processing all packets can require significant storage and computational resources, especially in high-traffic networks[11].

Flow-based monitoring aggregates packets into flows based on shared attributes within a defined time interval. Instead of analyzing individual packets, flow monitoring relies on summarized features such as packet counts, byte counts, and flow duration. This reduces data volume and improves scalability, although it provides less detailed visibility into traffic behavior [11].

Consequently, packet-level and flow-based monitoring represent different trade-offs between traffic visibility, detection capability, and resource efficiency.

2.3 Intrusion Detection in Wi-Fi Networks

Wireless Intrusion Detection Systems (WIDS) are used to monitor wireless network traffic and detect malicious activity [5]. They are typically used together with other security mechanisms, such as authentication and encryption, and provide an additional layer of protection[3].

In Wi-Fi networks, intrusion detection is often based on the analysis of network traffic and protocol behavior [4]. Two common approaches are signature-based detection and anomaly-based detection[4, 23]. Signature-based IDS compare observed traffic with known attack patterns stored in a database[4, 24]. This approach is effective for detecting known attacks, but it cannot detect new or unknown threats that do not match existing signatures[24]. Anomaly-based IDS instead model normal network behavior and detect deviations from this behavior as potential attacks[4, 24].

In many cases, anomaly-based systems use adaptive features such as packet rates, timing, or aggregated traffic values to define normal behavior and detect deviations [25]. This is important in network environments where monitoring and analysis introduce additional computational overhead.

2.4 Computational Cost, Energy Consumption, and Sustainability

The placement of intrusion detection components in a network can affect detection performance. In Wi-Fi environments, centralized IDS may experience reduced performance due to factors such as distance from the monitored traffic and frame loss. Distributed IDS place monitoring components closer to different parts of the network, which can improve detection accuracy and reduce information loss[26]

Real-time intrusion detection requires continuous processing of network traffic, introducing computational overhead and sustained resource usage [6]. The computational cost of an IDS depends on the detection method and system design. More advanced methods, such as machine learning-based approaches, require continuous data processing, which increases resource usage[3] Energy consumption in IDS is closely related to the amount of computation performed. Systems that require more processing power and memory generally consume more energy, while simpler methods can reduce energy usage[27]. This creates a trade-off between detection performance and energy efficiency[17]. In large-scale systems, energy consumption is also linked to carbon emissions, depending on the carbon intensity of the electricity used[18]. Metrics such as energy per inference can be used to evaluate the efficiency of IDS, especially in systems that process data continuously[28].

In information and communication technology (ICT), sustainability is often evaluated by analyzing the relationship between energy consumption and carbon emissions. The carbon footprint of a system is closely related to how much energy it uses, although this relationship is not always directly proportional due to differences in system efficiency and infrastructure[19].

Carbon emissions from computing systems can be estimated by combining total energy consumption with the carbon intensity of the electricity used. Carbon intensity represents the amount of greenhouse gas emissions per unit of electricity

This value varies depending on geographic location and time, since different energy sources are used in different regions and periods. As a result, the same computational task can produce different amounts of emissions depending on where and when it is executed[18].

3 | Related Work

3.1 IDS Evaluation and Sustainability Metrics

Research in cybersecurity shows that AI-based security systems can have a high environmental impact due to their computational requirements [16, 29]. Many intrusion detection systems (IDS) rely on machine learning and deep learning models, which require significant processing power and therefore increase energy consumption.

Despite this, IDS evaluation continues to focus primarily on detection-oriented metrics such as accuracy, precision, recall, and F1-score, while energy consumption and carbon emissions are less frequently incorporated into evaluation frameworks [17]. This reflects a broader tendency within machine learning research, where predictive performance is often prioritized over computational efficiency [14].

Some studies additionally include runtime-related metrics such as latency, throughput, computational overhead, CPU load, and memory usage to evaluate system performance in real-time environments [7, 9, 15]. However, these measurements are commonly analyzed separately from sustainability considerations.

To address this limitation, several studies propose integrating sustainability-oriented metrics into IDS evaluation. Metrics such as energy consumption, energy per inference, and eco-efficiency indices attempt to combine detection capability with resource efficiency [7, 14]. These approaches explicitly acknowledge the relationship between detection performance and computational demand.

Overall, prior research presents a fragmented evaluation landscape in which detection performance, runtime efficiency, and environmental impact are often evaluated independently rather than within a unified system-level framework.

3.2 Energy-Aware and Carbon-Aware IDS

Recent research introduces carbon-aware intrusion detection approaches in which energy consumption and carbon emissions are incorporated into system design and optimization objectives [17]. These approaches aim to balance detection performance with environmental impact.

Adaptive strategies, including reinforcement learning-based optimization, are used to dynamically adjust detection behavior according to both security and sustainability objectives [17]. This represents a shift from static IDS evaluation toward more adaptive and context-aware operation.

Carbon-aware computing research further demonstrates that environmental impact depends on when and where computation is performed because carbon intensity varies across

locations and time periods [18]. This enables systems to reduce emissions by scheduling computation during periods of lower carbon intensity or by relocating workloads geographically.

However, existing approaches primarily emphasize model-level optimization, while broader system-level interactions between deployment architecture, computational overhead, and sustainability receive comparatively less attention. In addition, consistent measurement and reporting of energy consumption and carbon emissions remain challenging across studies [30], which limits comparability and reproducibility.

3.3 IDS Deployment in Resource-Constrained and Distributed Environments

Prior research shows that IDS placement influences traffic visibility, detection latency, communication overhead, and resource utilization [26]. In distributed environments such as IoT and edge networks, IDS deployment must operate under constraints related to processing power, memory availability, and energy consumption [9].

In IoT systems, where devices are often resource-constrained, energy efficiency is a central concern. Green IoT research therefore focuses on reducing energy consumption while maintaining acceptable system performance [21]. Similarly, edge-based IDS deployments attempt to balance local processing capabilities with centralized coordination.

Machine learning-based IDS are increasingly used in these environments because they can improve detection capability beyond traditional signature-based methods. However, such approaches also introduce additional computational overhead due to continuous data processing and model execution [9]. Lightweight and ensemble-based approaches may reduce computational complexity, although they are frequently evaluated primarily from a detection-performance perspective rather than from a broader sustainability perspective [8].

Several studies rely on centralized evaluation setups that may not accurately reflect real-world distributed deployment conditions [8]. As a result, conclusions regarding scalability, resource efficiency, and deployment feasibility may be difficult to generalize across practical network architectures.

Network softwarization technologies such as Software-Defined Networking (SDN) and Network Function Virtualization (NFV) provide additional deployment flexibility through dynamic resource allocation and virtualized network functions [20]. These approaches further highlight the importance of considering system-level architecture when evaluating IDS performance and sustainability.

3.4 Research Gap and Thesis Positioning

The reviewed literature demonstrates increasing interest in sustainable and energy-aware intrusion detection. Existing studies investigate detection performance, lightweight IDS design, runtime efficiency, and carbon-aware optimization. However, these aspects are frequently analyzed separately rather than within a unified evaluation framework.

In particular, much of the current research focuses either on model-level optimization or on specific deployment domains such as IoT or data-center environments. Comparatively fewer studies explicitly evaluate sustainability-aware intrusion detection in Wi-Fi environments, especially in relation to access-point-level monitoring and distributed deployment

architectures.

Furthermore, the relationship between IDS placement, computational overhead, energy consumption, and estimated carbon emissions has only been partially addressed in existing research. Many evaluation approaches rely on simplified or centralized scenarios that may not accurately represent practical deployment conditions.

These limitations motivate the need for system-level evaluation approaches that jointly consider detection effectiveness, resource usage, energy consumption, and environmental impact in Wi-Fi intrusion detection systems.

Table 3.1: Summary of Related Work and Research Gap

Work	Focus Area	Energy	Security	Carbon Emission
[7]	Green IDS survey	✓	✓	×
[8]	Green IoT edge IDS	✓	✓	×
[9]	ML-driven IDS energy analysis	✓	✓	×
[14]	Eco-aware anomaly detection	✓	×	✓
[20]	Energy-efficient software networks	✓	×	×
[21]	Green IoT and energy efficiency	✓	×	✓
[26]	Distributed Wi-Fi IDS	×	✓	×
[30]	ML energy and carbon reporting	✓	×	✓
[31]	IDS in 5G and Wi-Fi networks	×	✓	×
This thesis	Wi-Fi IDS sustainability evaluation	✓	✓	✓

4 | Methodology

This chapter describes the methodology used to evaluate intrusion detection performance, network performance, energy consumption, and estimated carbon emissions in Wi-Fi networks. The study adopts a simulation-based experimental methodology implemented in ns-3. The methodology is designed to enable controlled and reproducible comparison of intrusion detection models under identical network conditions.

Real-world measurement of intrusion detection performance, energy consumption, and carbon emissions is difficult due to variability in traffic conditions, hardware differences, background network activity, and limited access to detailed energy data. Simulation enables precise control over parameters such as traffic rate, packet size, network topology, IDS configuration, and energy model settings. This makes it possible to isolate the effect of different intrusion detection strategies and compare their trade-offs in a consistent way.

In addition, the simulation-based approach allows fine-grained observation of system-level interactions between traffic behavior, intrusion detection logic, and resource consumption. This is particularly important in wireless environments, where communication overhead and processing overhead are closely connected. By integrating network simulation, IDS modeling, and energy estimation within the same framework, the methodology enables a unified evaluation of performance, efficiency, and sustainability.

The packet rate, measured as packets per second (PPS), is used as the primary detection feature. PPS directly reflects traffic intensity and is suitable for identifying high-rate attack behavior such as denial-of-service attacks. UDP is selected as the transport protocol because it does not include congestion control mechanisms. This allows traffic to be generated at a constant rate, making it possible to model both normal and attack traffic in a controlled and repeatable manner.

4.1 Research Design

This study uses a quantitative and experimental research design based on simulation. Simulation can be used to analyze system behavior and study how networks operate under different controlled conditions [32]. In this study, simulation is particularly suitable because detailed energy usage and IDS processing overhead are difficult to measure consistently in real systems.

Each simulation run is divided into two phases. During the first phase, only normal traffic is present. This phase is used as the baseline period. During the second phase, malicious traffic is introduced while normal traffic continues. This makes it possible to compare normal and abnormal behavior under identical network conditions. The structure also follows the principle of anomaly-based intrusion detection, where normal behavior is first observed and

deviations are later identified [33].

Each experiment follows the same procedure:

1. The simulation environment is initialized with the selected topology, traffic parameters, and energy model settings.
2. Normal traffic is generated and continues throughout the simulation.
3. Traffic statistics are recorded, with particular focus on PPS.
4. Depending on the IDS model, either a fixed threshold or a baseline-derived statistical threshold is defined.
5. Attack traffic is introduced during the second half of the simulation.
6. The IDS evaluates incoming traffic using the selected detection approach.
7. Detection results, network performance, and energy consumption are recorded.
8. The simulation is repeated using multiple random seeds.
9. Final results are calculated as average values across all runs.

Three IDS models are evaluated: a static threshold-based IDS, an adaptive anomaly-based IDS, and a hybrid IDS combining adaptive and signature-based detection. These models represent different levels of monitoring complexity and computational overhead.

The use of a controlled baseline phase followed by an attack phase ensures that detection performance can be evaluated relative to known normal behavior. This design reduces ambiguity in classification results and allows clear separation between false positives and true attack detections. Furthermore, repeating experiments using multiple random seeds reduces the impact of stochastic variation in packet scheduling and wireless channel behavior.

The choice of comparing static, adaptive, and hybrid IDS models reflects different design trade-offs commonly discussed in intrusion detection research. Static models represent low-cost baseline approaches, adaptive models reflect anomaly-based detection using statistical learning, and hybrid models combine multiple detection mechanisms to improve robustness. This categorization enables systematic evaluation of how increasing detection complexity affects both performance and resource usage.

The experimental design addresses the research questions as follows. RQ1 is addressed by measuring energy consumption at the AP and converting it into estimated carbon emissions using region-specific carbon intensity values. RQ2 is addressed by comparing IDS configurations deployed at the AP under identical traffic conditions. RQ3 is addressed by comparing IDS models with different computational complexity, which represents different levels of monitoring intensity and resource usage.

4.2 System Model

Network Model

The simulated network represents an infrastructure-based Wi-Fi topology. It consists of one access point (AP), five normal nodes, and one attacker node. The normal nodes generate legitimate traffic, while the attacker node generates malicious high-rate traffic during the attack phase.

All traffic is routed through the AP. Therefore, the AP acts as the centralized monitoring point where intrusion detection and energy measurement are performed. This reflects infrastructure-based wireless networks, where communication between stations is coordinated through a central device [34]. Centralized monitoring is also relevant in IoT and edge environments, where processing is often performed close to where data is generated. In such systems, APs and gateways are natural locations for detecting anomalous behavior and potential attacks [33].

Network Topology: Wireless IDS Simulation

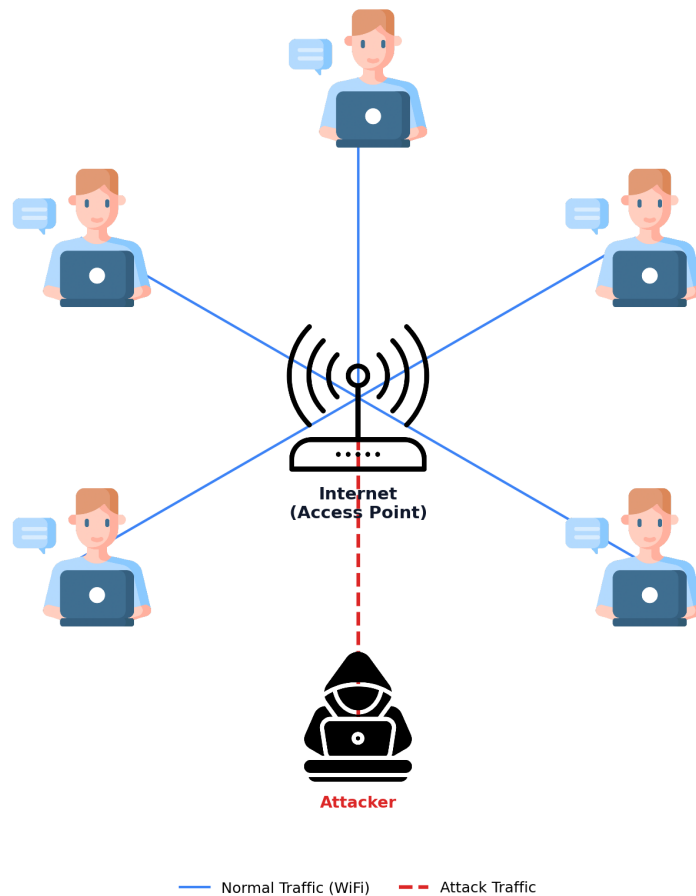


Figure 4.1: Simulated star topology used in the experiments. All traffic is routed through the AP, where intrusion detection and energy measurement are performed.

Traffic Model

The traffic model is designed to create controlled and repeatable conditions. Two types of traffic are used: normal traffic and attack traffic. Normal traffic represents regular communication from legitimate nodes, while attack traffic represents abnormal high-rate communication.

UDP is used as the transport protocol. Since UDP does not adjust its sending rate based on congestion, it allows traffic to be generated at a steady rate. This makes it suitable for modeling high traffic loads such as denial-of-service attacks [34]. DoS attacks aim to overload the system by generating large amounts of traffic, which can reduce network performance and disrupt legitimate communication [33].

The packet rate depends on the data rate and packet size. PPS is calculated as:

$$\text{PPS} = \frac{R}{8L}$$

where R is the data rate in bits per second and L is the packet size in bytes. Smaller packets or higher data rates increase PPS, which increases the load on the network and the IDS.

From a detection perspective, PPS is particularly suitable as a feature because it is independent of packet payload and can be observed even when traffic is encrypted. This makes it relevant for modern wireless networks where encryption limits the applicability of deep packet inspection. High PPS values typically indicate abnormal traffic bursts, which are characteristic of volumetric attacks such as denial-of-service.

Using PPS as a single aggregated feature also reduces computational complexity compared to feature-rich IDS approaches. This aligns with the study's focus on evaluating trade-offs between detection performance and resource efficiency.

4.3 Simulation Environment and Implementation

The simulation framework is implemented in ns-3. The network is configured as an infrastructure-based Wi-Fi system where the AP acts as the central communication node. Legitimate nodes generate normal UDP traffic, while the attacker node generates malicious UDP traffic during the attack phase.

The IEEE 802.11ax standard introduces Orthogonal Frequency Division Multiple Access (OFDMA), which allows multiple users to transmit simultaneously by dividing the channel into smaller resource units [35]. This improves network efficiency and reduces contention in dense environments [35, 36]. In this study, the wireless configuration and node setup are kept fixed to ensure controlled and comparable experimental conditions.

The three IDS configurations are implemented as separate ns-3 programs: static IDS, adaptive IDS, and hybrid IDS. Each configuration uses the same topology, simulation duration, traffic parameters, packet size, and energy settings. This ensures that differences in the results are caused by the IDS design rather than changes in the simulation environment.

FlowMonitor is used to collect flow-level statistics during each simulation run. The collected statistics are exported as XML files and processed after the simulation. Normal and attack traffic are separated using destination ports, where one port corresponds to normal traffic and another to attack traffic. This enables separate analysis of legitimate and mali-

cious flows.

To automate the experiments, Python scripts execute the ns-3 simulations multiple times using different random seeds. Each script runs the corresponding ns-3 program through the command line, captures the simulation output, parses FlowMonitor XML files, extracts relevant metrics, and aggregates the results across runs. This creates a consistent and reproducible evaluation pipeline.

The separation of simulation execution and post-processing ensures modularity in the experimental framework. The ns-3 simulations are responsible for generating network behavior and raw data, while the Python-based analysis pipeline performs metric extraction, aggregation, and transformation. This separation reduces implementation bias and ensures that identical processing steps are applied across all IDS configurations.

Furthermore, using a shared analysis pipeline minimizes inconsistencies that could otherwise arise from model-specific processing scripts. This design choice improves internal validity and ensures that observed differences between IDS models are attributable to the models themselves rather than to differences in data processing.

Nr.	Category	Tool
1	Hardware Resources	Computer for simulation
2	Software Resources	Linux OS
		ns-3
3	Programming	C++
		Python

Table 4.1: Resources and tools used in the simulation experiments.

Nr.	Parameter	Value
1	Normal traffic data rate	1 Mbps
2	Attack traffic data rate	20 Mbps
3	Normal packet size	1024 bytes
4	Attack packet size	1024 bytes
5	Baseline duration	10 s
6	Attack duration	10 s
7	Constant k	3
8	Initial energy	100 J
9	TxCURRENT	20 mA
10	RxCURRENT	22 mA
11	IdleCurrent	0.5 mA

Table 4.2: Simulation parameters used in the experiments.

4.4 Experimental Scenario

Each simulation run has a fixed duration of 20 seconds. During the first 10 seconds, only normal traffic is generated. This period represents the baseline phase and is used to observe normal network behavior. During the second 10 seconds, attack traffic is introduced while normal traffic continues. This period represents the attack phase.

The same 20-second scenario is used for all IDS models. Each model is evaluated under identical traffic conditions and repeated across multiple random seeds. This enables controlled comparison of detection performance, network performance, energy consumption, and estimated carbon emissions.

The simulation parameters are shown in Table 4.2.

4.5 Experimental Variables

The independent variables are the IDS model type and attack traffic configuration:

$$M \in \{\text{static, adaptive, hybrid}\}$$

$$T = (r_a, N_a)$$

where r_a denotes the attacker data rate and N_a denotes the number of attacker nodes. The dependent variables are grouped into three categories:

$$D = (\text{precision, recall, F1-score})$$

$$N = (\text{throughput, delay})$$

$$S = (\text{energy consumption, carbon emissions})$$

where D represents detection performance, N represents network performance, and S represents sustainability metrics.

The selected independent variables reflect both detection logic and traffic conditions. The IDS model type captures differences in detection strategy and computational complexity, while the traffic configuration controls the intensity and scale of the attack. Together, these variables allow analysis of how detection performance and resource consumption scale with increasing network load.

The dependent variables are structured to capture three complementary perspectives: security effectiveness, network behavior, and environmental impact. This multi-dimensional evaluation is necessary to analyze trade-offs between detection accuracy, system performance, and sustainability.

4.6 Intrusion Detection Models

The IDS is placed at the AP and continuously monitors incoming traffic. Detection is based on the observed PPS. Three IDS approaches are implemented and compared: static IDS, adaptive IDS, and hybrid IDS.

Each IDS model introduces computational overhead, modeled as energy consumption proportional to the number of processed packets.

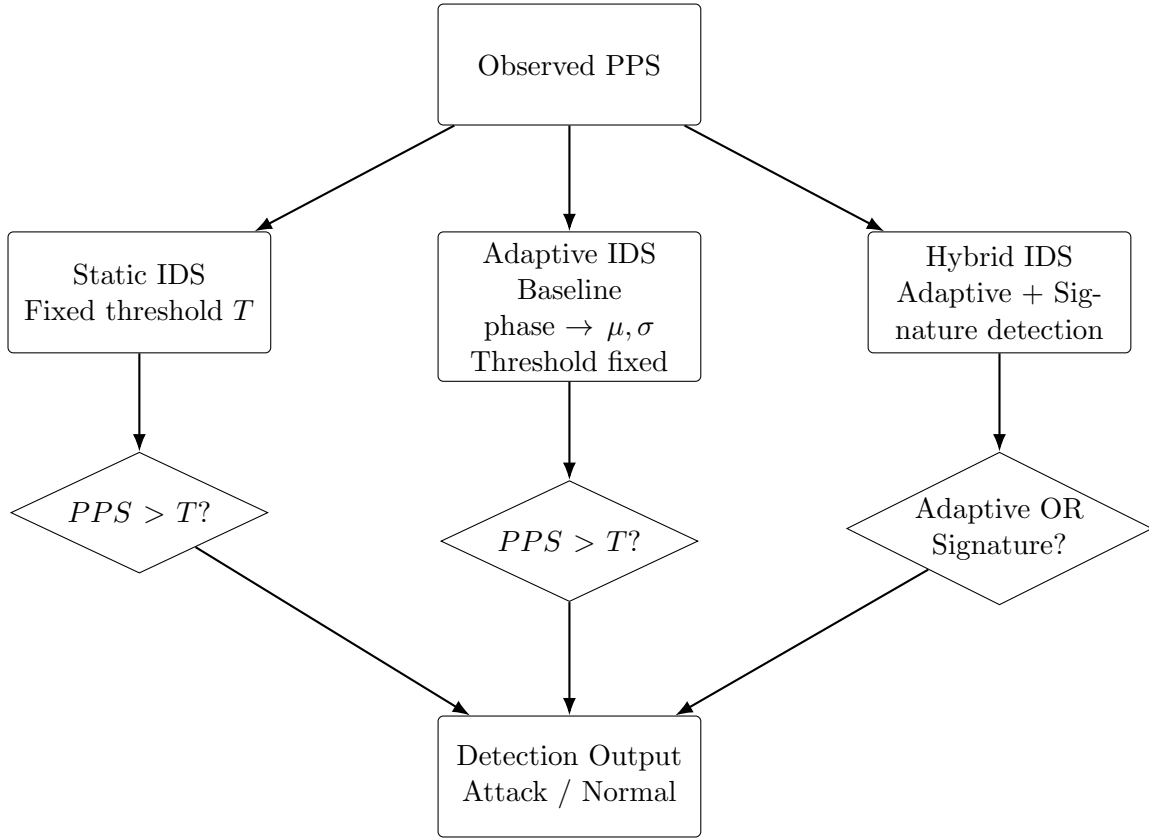


Figure 4.2: Detection logic for the three IDS models. The static IDS uses a fixed threshold, the adaptive IDS derives its threshold from baseline traffic and keeps it fixed during detection, and the hybrid IDS combines adaptive detection with signature-based detection using a logical OR condition.

Static IDS

The static IDS uses a predefined threshold. Traffic is classified as anomalous when the observed PPS exceeds this fixed threshold. This model does not adapt to traffic conditions and is used as a simple baseline.

The detection rule is:

$$\hat{y}(t) = \begin{cases} 1, & \text{if } PPS(t) > T_{\text{static}} \\ 0, & \text{otherwise} \end{cases}$$

where $\hat{y}(t) = 1$ indicates attack traffic and $\hat{y}(t) = 0$ indicates normal traffic.

Adaptive IDS

The adaptive IDS follows an anomaly-based approach. During the baseline phase, the IDS collects PPS values representing normal traffic behavior. These values are used to compute the mean and standard deviation:

$$\mu_{PPS} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sigma_{PPS} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu_{PPS})^2}$$

The detection threshold is then defined as:

$$T_{\text{adaptive}} = \mu_{PPS} + k \cdot \sigma_{PPS}$$

where $k = 3$. After the baseline phase, the threshold is fixed and used for the remainder of the simulation:

$$\hat{y}(t) = \begin{cases} 1, & \text{if } PPS(t) > \mu_{PPS} + k\sigma_{PPS} \\ 0, & \text{otherwise} \end{cases}$$

The adaptive IDS assumes that normal traffic during the baseline phase provides a representative estimate of expected traffic behavior. Deviations from this distribution are interpreted as anomalies.

Hybrid IDS

The hybrid IDS combines adaptive anomaly detection with signature-based detection. The adaptive component uses the statistical threshold described above. The signature-based component detects sustained high-rate traffic based on recent PPS values. The signature is defined as a sequence of consecutive high PPS values, representing sustained traffic bursts typical of denial-of-service behavior.

The final detection decision is based on a logical OR condition:

$$\hat{y}(t) = f_{\text{adaptive}}(t) \vee f_{\text{signature}}(t)$$

where $f_{\text{adaptive}}(t)$ represents threshold-based anomaly detection and $f_{\text{signature}}(t)$ represents detection of predefined traffic patterns.

The signature component is defined as:

$$f_{\text{signature}}(t) = \begin{cases} 1, & \text{if } PPS(t), PPS(t-1), \dots, PPS(t-n) > T \\ 0, & \text{otherwise} \end{cases}$$

This allows the hybrid IDS to detect both sudden deviations from normal traffic and persistent attack patterns.

4.7 Energy and Carbon Modeling

Energy consumption is measured at the AP, where all traffic is processed and intrusion detection is performed. The energy model includes both communication-related energy and IDS processing overhead.

A radio energy model is used to capture energy consumption across different device states, such as transmission, reception, idle, and sleep. Each state has a different current or power consumption level. The total radio energy depends on the time the AP spends in each state during the simulation. Increased traffic may increase time spent in active states such as transmission and reception, thereby increasing total energy consumption [37].

The radio energy consumption is calculated as:

$$E_{\text{radio}} = E_{\text{initial}} - E_{\text{remaining}}$$

Each IDS model also introduces an abstract processing cost proportional to the number of processed packets. The total energy consumption is calculated as:

$$E_{\text{total}} = E_{\text{radio}} + E_{\text{IDS}}$$

where E_{IDS} represents IDS-specific computational overhead. In the static IDS, this overhead is modeled as a fixed cost per processed packet. In the adaptive IDS, the overhead is higher due to the maintenance of traffic statistics. In the hybrid IDS, the overhead includes both adaptive processing and signature matching.

The inclusion of IDS processing overhead as a separate energy component reflects the fact that intrusion detection is not only a network monitoring task but also a computational process. By modeling IDS overhead as proportional to the number of processed packets, the methodology captures the relationship between traffic volume and processing cost.

This abstraction enables comparison between IDS models with different computational characteristics without relying on hardware-specific measurements. Although simplified, this approach is consistent with common practices in simulation-based energy analysis and provides meaningful insights into relative energy efficiency.

The energy model is an abstraction and does not represent exact hardware measurements. However, it provides a controlled approximation of relative energy usage across different IDS configurations.

To estimate environmental impact, total energy consumption is converted into estimated carbon emissions. Energy is first converted from joules to kWh. Carbon emissions are then calculated as:

$$CO_2 = E_{\text{total}} \cdot CI$$

where CO_2 represents estimated carbon emissions in gCO₂, E_{total} is total energy consumption in kWh, and CI is carbon intensity in gCO₂/kWh.

Carbon intensity data is obtained from Electricity Maps [38]. In this study, carbon data is not used to control the system during runtime. Instead, it is used during post-processing to estimate the environmental impact of the measured energy consumption.

Carbon emissions are evaluated for Sweden, India, the USA, Australia, Germany, and Tanzania. These regions are selected to represent diverse electricity generation profiles with varying carbon intensities. The selection includes both low-carbon energy systems and higher-carbon-intensity systems, as well as large industrialized economies. This multi-region evaluation addresses a limitation in existing research, where carbon impact is often assessed using single-region assumptions or omitted entirely. By incorporating multiple energy mixes, the study enables a more realistic system-level analysis of how identical IDS configurations can result in different environmental impacts depending on deployment context.

It is important to note that the carbon emission model assumes a linear relationship between energy consumption and emissions, which is a common approximation in ICT sustainability studies. While real-world emissions may vary due to temporal fluctuations in

energy production, the use of fixed regional carbon intensity values allows consistent comparison across IDS models.

By decoupling energy measurement from carbon evaluation, the methodology enables reuse of energy results under different environmental assumptions. This makes it possible to extend the analysis to additional regions or future energy scenarios without modifying the underlying simulation.

4.8 Performance Metrics

The evaluation uses three categories of metrics: detection performance, network performance, and sustainability metrics.

Detection Performance Metrics

Detection performance is evaluated using confusion matrix values: true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN). These values are obtained from the simulation output and used to calculate precision, recall, and F1-score:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Network Performance Metrics

Network performance metrics are obtained using FlowMonitor. During each simulation run, flow-level statistics are recorded and exported as XML files. These files are processed using Python to extract transmitted packets, received packets, total received bytes, and accumulated delay.

Throughput is computed based on the total number of received bytes during a simulation run:

$$\tau = \frac{8B}{10^6}$$

where B denotes the total number of received bytes. This formulation represents the total volume of successfully delivered data per simulation run rather than a time-normalized transmission rate. Since all simulation runs are executed with identical duration and traffic conditions, the throughput values remain comparable across IDS configurations.

Packet delivery ratio is calculated as:

$$\text{PDR} = \frac{P_{\text{received}}}{P_{\text{sent}}}$$

The packet loss rate is calculated as:

$$\text{Loss Rate} = 1 - \text{PDR}$$

Average delay is calculated as:

$$d = \frac{D_{\text{total}}}{P_{\text{received}}}$$

Sustainability Metrics

Sustainability is evaluated using total energy consumption and estimated carbon emissions. Energy consumption is extracted from the ns-3 simulation output for each run. Carbon emissions are calculated during post-processing by combining measured energy consumption with regional carbon intensity values.

4.9 Data Collection and Processing Pipeline

Data collection is performed during simulation runtime and consists of three categories: (i) security data, (ii) network performance data, and (iii) sustainability data.

Security data includes PPS values and detection outcomes produced by the IDS. Network performance data includes transmitted packets, received packets, throughput, and delay. Sustainability data includes energy consumption measured at the AP and derived carbon emissions.

The collected data follows a structured processing pipeline. First, raw simulation outputs and FlowMonitor XML files are generated. Second, Python scripts parse these files to extract relevant metrics. Third, derived metrics such as throughput, delay, precision, recall, F1-score, and carbon emissions are computed. Finally, results are aggregated across multiple simulation runs and used for analysis and visualization.

This structured pipeline ensures traceability from raw simulation data to final results and supports reproducibility of the experimental process.

4.10 Experiment Execution and Data Analysis

The simulation experiments are executed using an automated Python-based framework. Each IDS model is implemented as a separate ns-3 program and executed independently under identical simulation conditions.

For each simulation run, the framework performs the following steps:

1. execute the selected ns-3 IDS simulation,
2. capture simulation output,
3. collect FlowMonitor XML files,
4. extract detection, network, and energy metrics,
5. compute carbon emissions for each region,
6. aggregate results across random seeds.

All simulation outputs and FlowMonitor files are processed using a shared Python analysis pipeline. This ensures that parsing, metric computation, aggregation, and visualization

are performed consistently across all IDS models. By using a unified processing approach, inconsistencies between model-specific scripts are avoided.

The final reported values represent averages across multiple simulation runs. This reduces the influence of random variation between individual runs and improves robustness of the comparison between IDS configurations.

The analysis focuses on the relationship between detection performance, network performance, energy consumption, and estimated carbon emissions. This enables evaluation of how different IDS designs affect both security effectiveness and sustainability. In particular, the analysis examines whether increased detection complexity leads to improved detection performance, higher energy consumption, or different carbon impacts depending on the regional electricity mix.

4.11 Result Visualization

The experimental results are visualized using Python and Matplotlib after all simulation runs are completed and results are aggregated. For each IDS model, plots are generated to analyze throughput, delay, energy consumption, IDS performance, and carbon emissions.

Carbon emissions are visualized for Sweden, India, the USA, Australia, Germany, and Tanzania. This enables analysis of how identical energy consumption can produce different environmental impacts depending on the electricity mix.

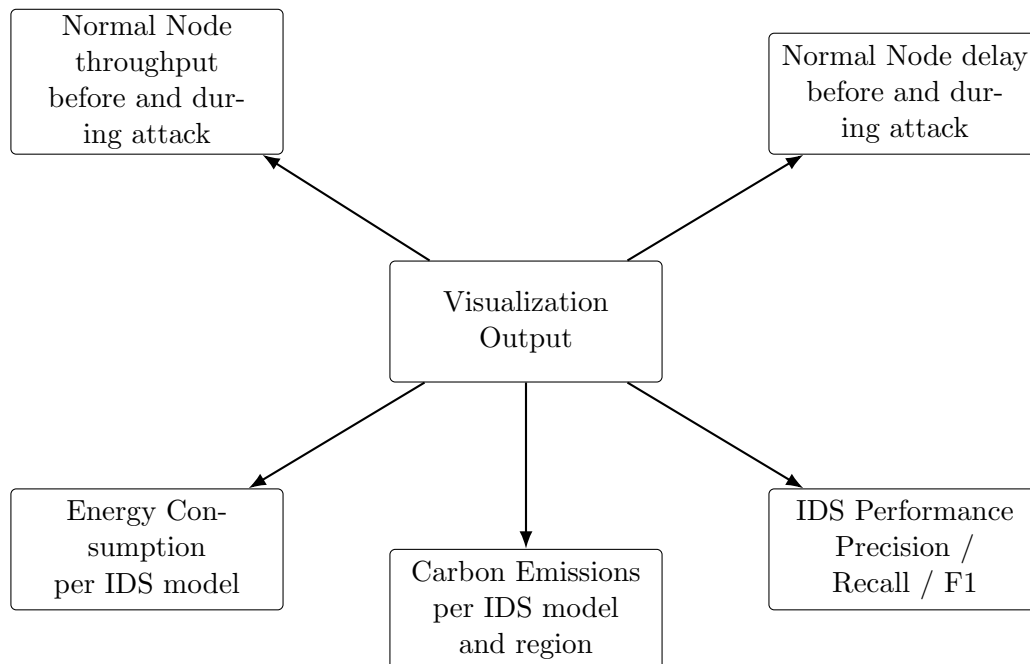


Figure 4.3: Overview of the main result visualizations used in the study.

4.12 Validity and Limitations

The simulation-based methodology provides strong control over experimental conditions and enables reproducible comparison between IDS models. However, several limitations must be acknowledged.

First, the results depend on the accuracy of the simulation models and may not fully represent real-world Wi-Fi behavior. Second, the topology is simplified and static, which improves experimental control but reduces realism. Third, UDP traffic enables controlled high-rate traffic generation but does not represent congestion-aware transport behavior. Fourth, the energy model is an abstraction based on predefined parameters and does not represent exact hardware measurements. Fifth, carbon emissions are estimated using fixed regional carbon intensity values and do not reflect temporal variation in electricity production.

Internal validity is supported by the use of identical simulation conditions across all IDS models, controlled traffic generation, and consistent data processing. External validity is limited due to the simplified topology and traffic model, which may not fully capture real-world variability in wireless networks.

Construct validity is addressed by selecting metrics that directly correspond to the research objectives, including detection performance, network performance, energy consumption, and carbon emissions. However, the use of PPS as the sole detection feature may limit generalizability to more complex attack scenarios.

Despite these limitations, the methodology provides a controlled and systematic framework for analyzing trade-offs between intrusion detection performance and sustainability in wireless networks.

5 | Results

5.1 Comparison of IDS Models

This chapter presents the experimental results obtained from the simulations of the Static, Adaptive, and Hybrid IDS models. The results from each simulation run were stored and subsequently aggregated into combined result files. The comparison focuses on energy consumption analysis, carbon emissions analysis, network performance analysis, detection performance, and detection efficiency.

5.1.1 Energy Consumption Analysis

Figure 5.1 displays the distribution of energy consumption among the three IDS approaches. It is evident that all three approaches have very close energy consumption values, although some minor differences exist between the three models. Static IDS records the least median value for energy consumption while the Adaptive IDS follows closely behind.

The Hybrid IDS consumes more energy than any other approach, although the difference between them is not significant (at best on the order of 10^{-7} kWh). Although it is small, this difference exists among all runs, suggesting that IDS design has an effect on energy consumption.

5.1.2 Carbon Emission Analysis

The estimated carbon emissions from each IDS approach across regions can be seen in Figure 5.2. As anticipated, the carbon emissions patterns correlate strongly with the energy consumption patterns, but here Hybrid IDS has a little more emissions and Static IDS a little less. One major thing to observe here is that there is a great difference in the amount of carbon emissions in different regions.

Emissions, for instance, would be very high in a place like India when compared with the case of Sweden, even though the system works in the same way. The lesson here is that IDS implementation can greatly vary in environmental terms depending on the energy mix of the region, not on whether it is hybrid IDS or static IDS that is used.

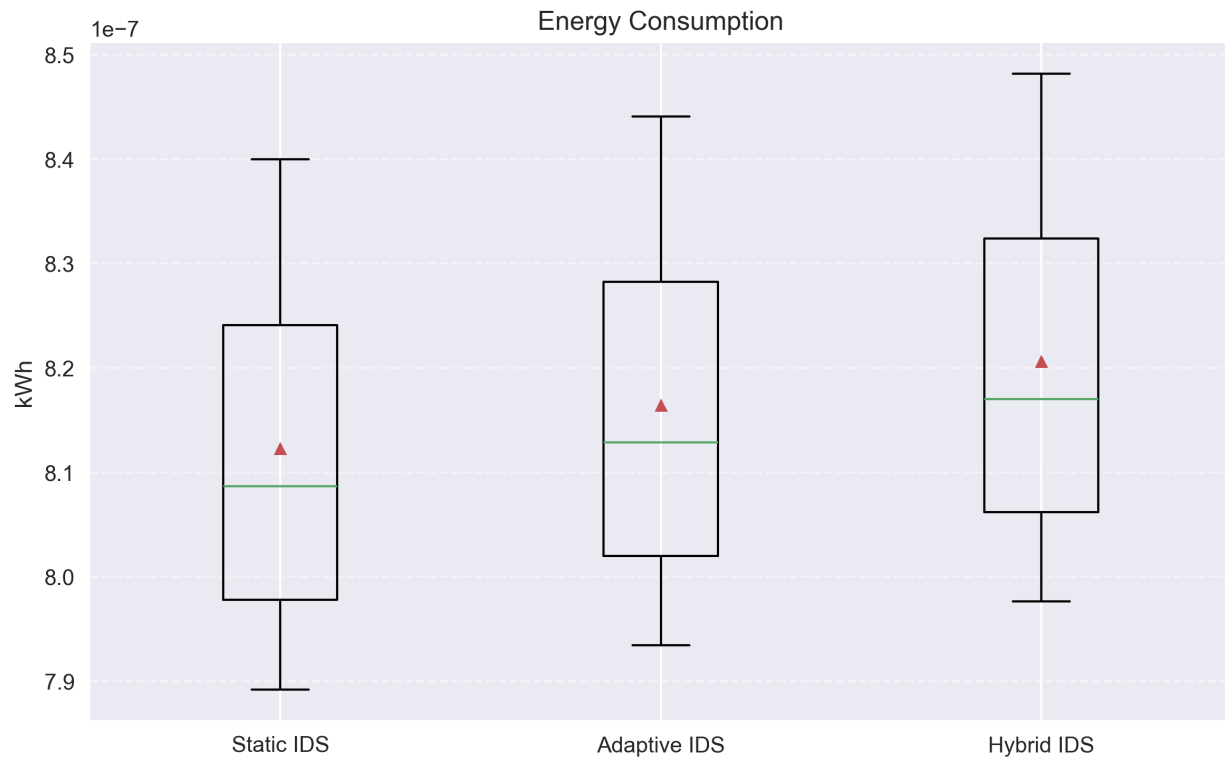


Figure 5.1: Energy consumption distribution across IDS models.

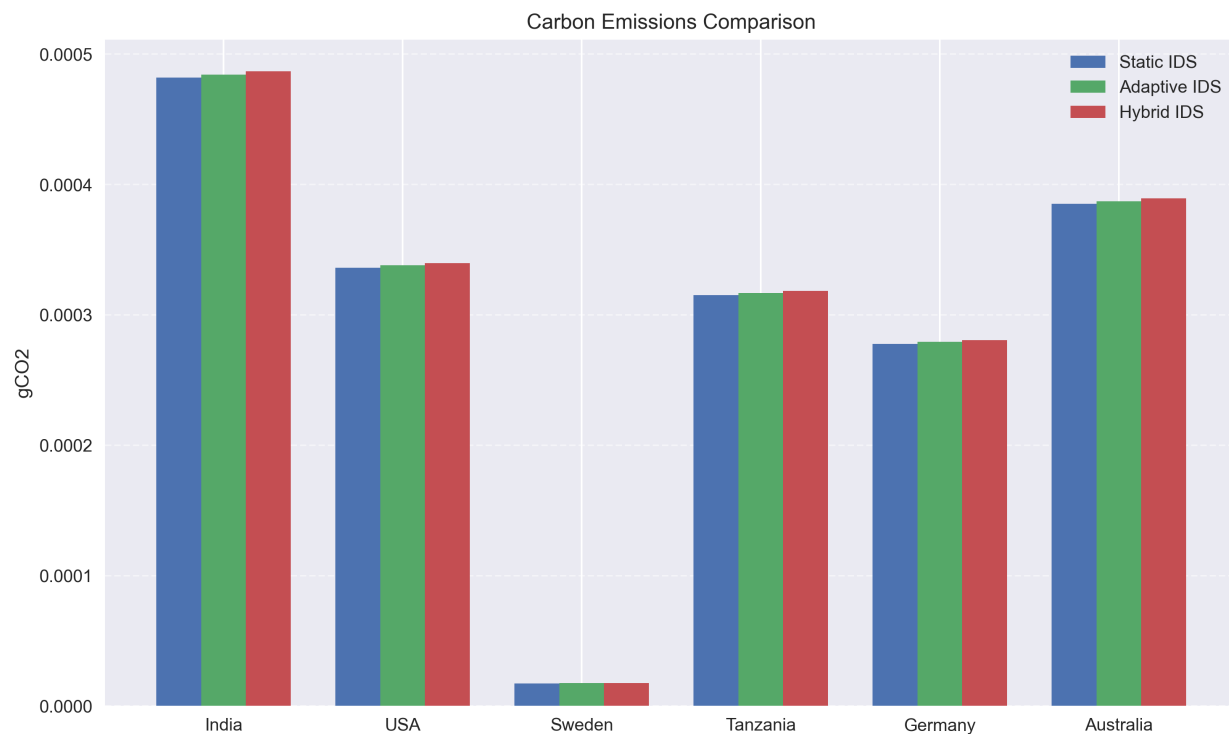


Figure 5.2: Carbon emissions across regions for different IDS strategies.

5.1.3 Network Performance Impact

Figure 5.3 and Figure 5.4 represent network performance when baseline and abnormal traffic occurs. From the results obtained, throughput is almost constant except for a minimal drop when abnormal traffic is considered for all the IDS strategies considered. Moreover, there is also an increase in delay under abnormal traffic conditions; however, this increase is the same regardless of the IDS strategy adopted.

It should be noted here that there were no significant differences observed in the throughput and delay for all the IDS strategies under consideration. This shows that IDS mechanism introduced no significant network performance overhead. The average throughput drops from 5.13376 Mbps to 5.11196 Mbps when under attack, representing a drop of about 0.42%. This drop suggests that the attack has an effect on the throughput performance but to a minimal level. There was no considerable difference observed among all the IDS models considered here

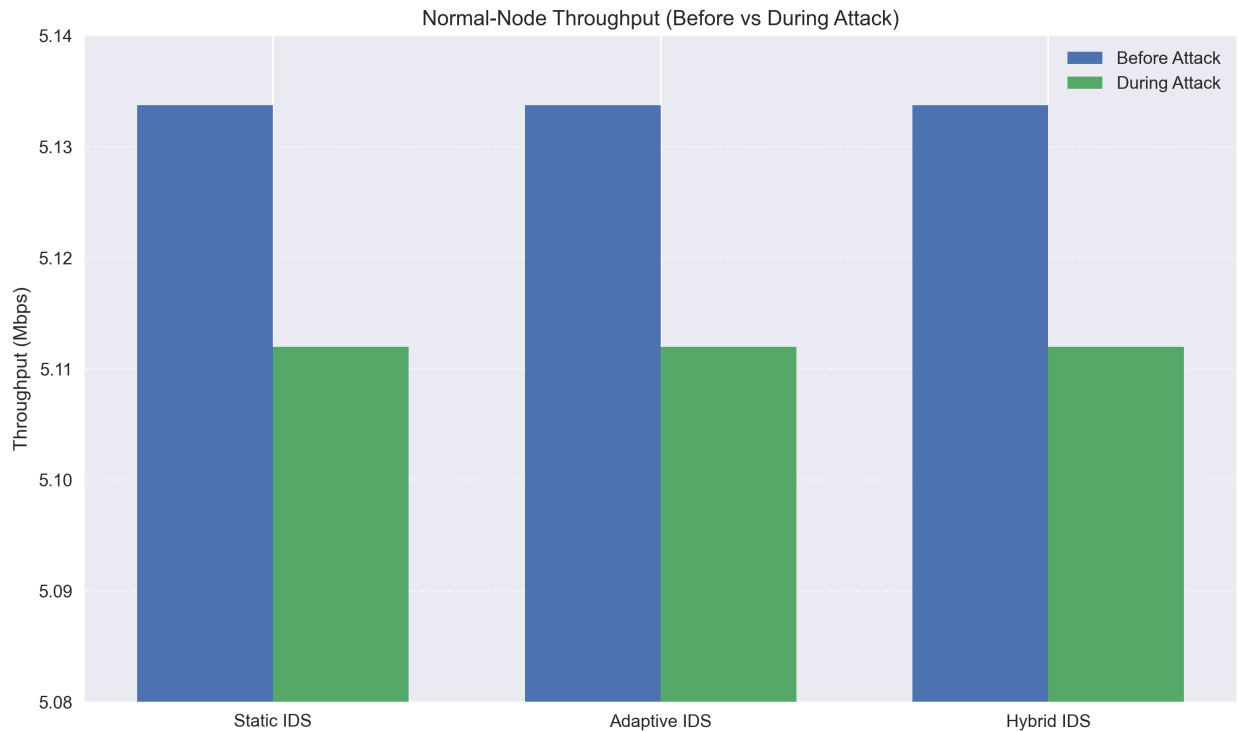


Figure 5.3: Throughput under baseline and abnormal traffic conditions.

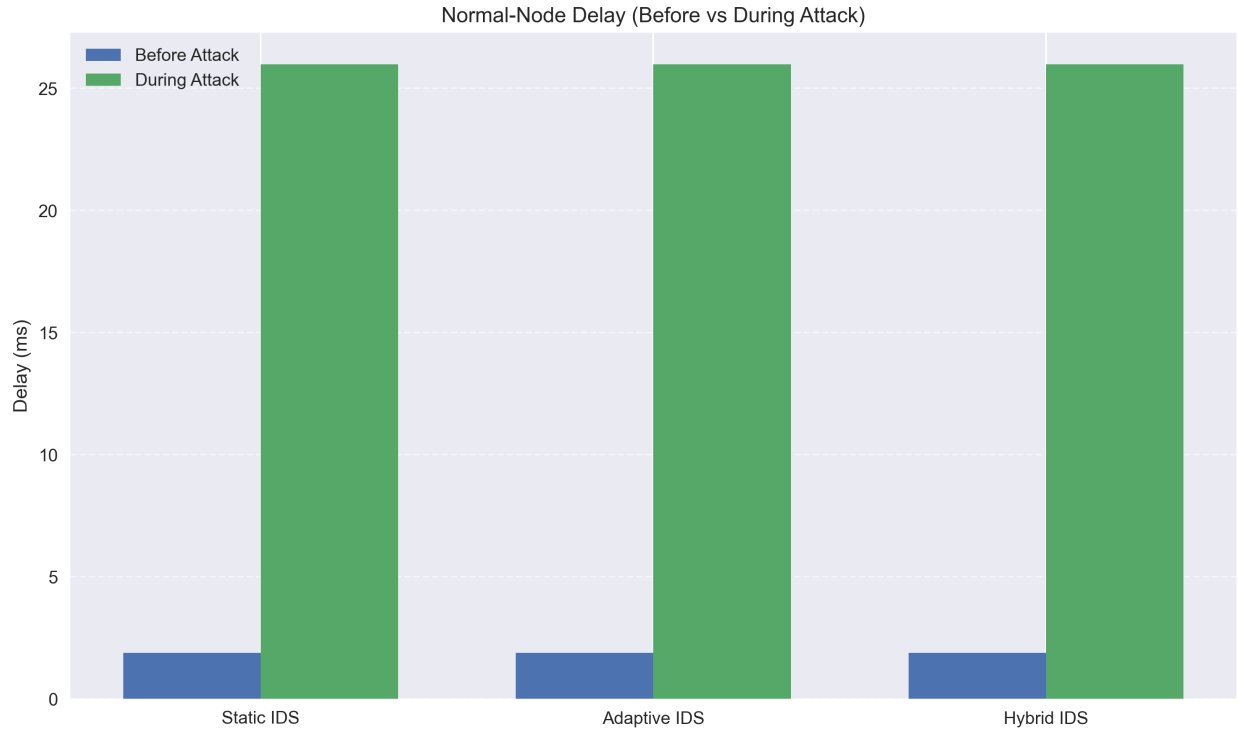


Figure 5.4: Average delay under baseline and abnormal traffic conditions.

5.1.4 Detection Performance

Figure 5.5 illustrates the accuracy in detection for the three IDS methods based on precision, recall, and F1 score. While the Static IDS method has perfect recall but relatively low precision, suggesting a high rate of false alarms, the Adaptive IDS method has perfect precision and high recall, making its F1 score impressive.

The Hybrid IDS method, on the other hand, boasts both a high precision and perfect recall, which makes its F1 score the highest among the three methods. However, the difference between the F1 scores of the Adaptive and Hybrid IDS methods is not significant, implying that there was not much improvement gained from integrating both detection mechanisms. Illustrates the accuracy in detection for the three IDS methods based on precision, recall, and F1 score.

5.1.5 Detection Efficiency

Figure 5.6 depicts the detection efficiency as F1-score with respect to energy consumption per unit. Such an efficiency indicator considers the detection process and the associated energy costs involved, thus giving an environmentally-friendly comparison of the IDS approaches under consideration.

From the findings presented below, it is evident that the efficiency of detection is not necessarily increased with improved detection. Despite having the highest detection score, the Hybrid IDS model consumes more energy. Consequently, the Adaptive IDS approach strikes a good balance of detection efficiency and energy consumption.

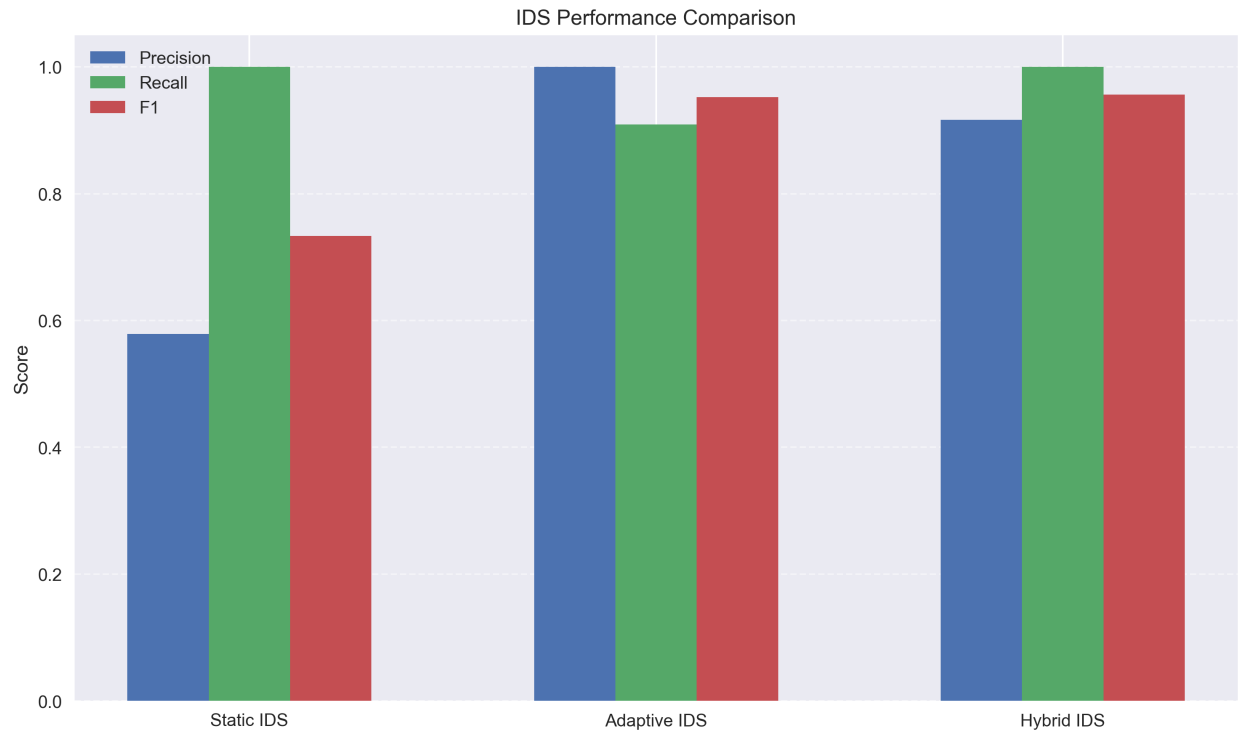


Figure 5.5: IDS performance comparison across models.

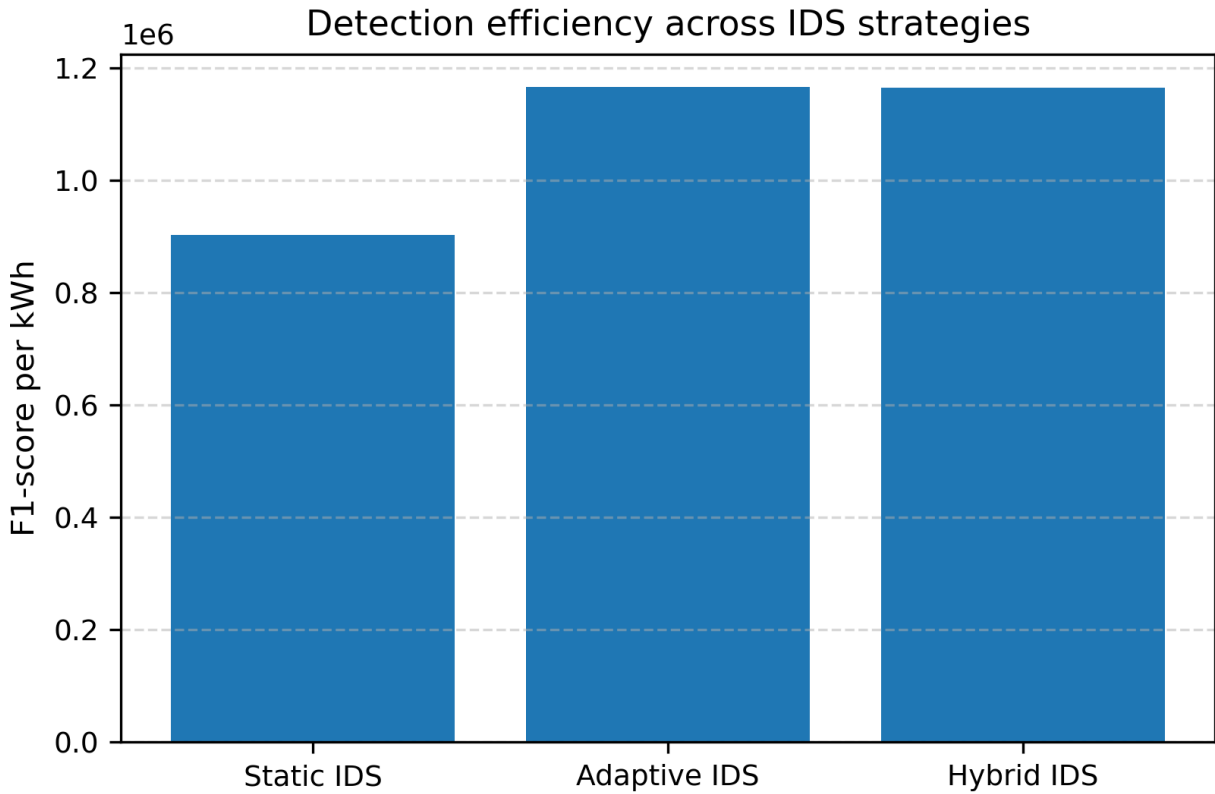


Figure 5.6: Detection efficiency (F1-score per kWh) across IDS strategies.

6 | Discussion and Analysis

This chapter analyses and discusses the results in relation to the research questions and relevant theory. The focus is on the trade-offs between detection performance, energy use, and environmental impact, with particular focus on the differences between the Static, Adaptive, and Hybrid IDS models.

6.1 Experimental Result Analysis

The energy consumption results show a consistent pattern across the three IDS models. The Static IDS consumes the least energy, followed by the Adaptive IDS, while the Hybrid IDS consumes the most energy. The small standard deviation ($\approx 1.78 \times 10^{-8}$ kWh) indicates stable behavior across simulation runs. The relative differences between the models are small but systematic. The Hybrid IDS consumes approximately 0.51% more energy than the Adaptive IDS and approximately 1.03% more than the Static IDS. This indicates that increased IDS complexity introduces additional energy cost, even when the absolute energy values are small.

The carbon emission results follow the same ranking as the energy consumption results. The Hybrid IDS produces the highest estimated emissions, followed by the Adaptive IDS, while the Static IDS produces the lowest emissions. This pattern directly reflects the energy consumption results, since emissions are derived from energy use and regional carbon intensity. The regional comparison shows that the same IDS energy consumption can result in different carbon emission values depending on the electricity mix used in the calculation. This means that carbon impact depends not only on IDS design, but also on deployment context.

The throughput results show a decrease from 5.13376 Mbps before the attack to 5.11196 Mbps during the attack, corresponding to a reduction of approximately 0.42%. This reduction is small compared to the delay increase. The delay results show a larger performance impact. The mean delay increases from 1.8873 ms to 25.98 ms during the attack, corresponding to an increase of approximately 13.8 times. Since the delay values are identical across all IDS models, the increase is caused by attack traffic rather than differences in IDS model complexity. This means that the main trade-off in this simulation is between detection performance and energy use, not between IDS complexity and network quality.

The Static IDS achieves perfect recall (1.0000), meaning that it detects all attack cases, but its precision is low (0.5789). This results in an F1-score of 0.7333 and indicates a high number of false positives. The Adaptive IDS achieves perfect precision (1.0000), recall of 0.9091, and an F1-score of 0.9524. This means that the Adaptive IDS avoids false positives but misses some attacks. The Hybrid IDS achieves precision of 0.9167, perfect recall (1.0000),

and the highest F1-score (0.9565). This represents the best overall detection performance. However, the improvement over the Adaptive IDS is marginal.

The detection efficiency results show that the Adaptive and Hybrid IDS strategies achieve nearly identical efficiency, while the Static IDS performs noticeably worse. This indicates that the lower precision of the Static IDS reduces its effectiveness relative to the energy consumed. In contrast, both Adaptive and Hybrid strategies provide a better balance between detection capability and energy usage. Given the marginal performance improvement of the Hybrid IDS over the Adaptive IDS, the results suggest that the Adaptive approach offers a competitive balance between efficiency and detection performance.

6.2 Detection Performance, Energy Use, and IDS Trade-offs

The results show a clear trade-off between detection performance and energy use across the three IDS models. The Hybrid IDS achieved the highest detection performance, followed closely by the Adaptive IDS, while the Static IDS showed the lowest overall performance due to a high number of false positives.

In terms of energy consumption, the Static IDS was the most energy-efficient model, the Adaptive IDS consumed moderately more energy, and the Hybrid IDS had the highest energy consumption. Quantitatively, the Hybrid IDS used approximately 0.51% more energy than the Adaptive IDS and approximately 1.03% more than the Static IDS. When comparing performance gains, the Adaptive IDS achieved detection performance close to the Hybrid IDS, while reducing energy consumption by approximately 0.51% relative to the Hybrid model.

This trade-off is consistent with earlier research showing that more complex intrusion detection methods require greater computational resources, leading to increased energy consumption [9, 13]. IDS systems are known to consume significant resources when processing large volumes of network traffic or applying advanced detection techniques [7]. Energy consumption in IDS arises from both computational processing and communication overhead, both of which scale with system complexity [7]. This explains the higher energy consumption observed in the Hybrid IDS, which combines multiple detection mechanisms.

A comparative interpretation of the results highlights three distinct operating points: (i) the Static IDS represents a low-energy, low-precision configuration, (ii) the Hybrid IDS represents a high-performance, high-energy configuration, and (iii) the Adaptive IDS lies between these, offering a near-optimal trade-off.

Overall, the relatively small increase in detection performance from Adaptive to Hybrid, compared to the additional energy cost, suggests diminishing returns at higher levels of model complexity. This indicates that the Adaptive IDS may represent a more efficient design choice in scenarios where both performance and energy consumption are important.

The results can also be interpreted using a simplified analytical model:

$$(D, E, C) = f(M, A, T)$$

where detection performance (D), energy consumption (E), and carbon emissions (C) depend on IDS model complexity (M), monitoring architecture (A), and traffic conditions

(T).

This reflects that increasing model complexity improves detection performance but also increases computational requirements and energy consumption [7]. The efficiency results further show that the Hybrid and Adaptive IDS achieve similar performance per unit of energy, while the Static IDS is less efficient due to lower detection performance.

6.3 Research Question Analysis

The results clearly show that energy use and carbon emissions change depending on the IDS model. The Static IDS uses the least energy, followed by the Adaptive IDS, while the Hybrid IDS uses the most. Since carbon emissions are calculated from energy use using regional carbon intensity values, the same pattern appears in the emission results. Even though the energy values are small, they become important when systems run continuously. IDS systems usually run all the time [3], so energy use builds up over time.

The results show that IDS design affects both energy use and carbon emissions. More complex IDS models increase energy use and emissions, while simpler models reduce energy use but also reduce detection performance.

The study looks at intrusion detection in a centralized system at the access point. In this setup, detection performance and energy use mainly depend on how complex the IDS model is. Earlier research shows that system design is also important for efficiency. IDS can be centralized, distributed, or hybrid, and each option has different trade-offs in computation and communication [7]. Centralized systems process all traffic in one place, which increases computation. Distributed systems process data closer to where it is created, which reduces communication but may increase local processing.

However, this study only looks at a centralized system and does not compare different architectures. Therefore, it does not give a full answer to how system design affects energy use and detection performance. The results also show that in a centralized system, IDS performance and energy use mainly depend on model complexity.

The results show that adaptive monitoring can reduce energy use while keeping high detection performance. The Adaptive IDS achieved detection performance close to the Hybrid IDS while reducing energy use by about 0.51%. This agrees with research on green IDS, where methods like adaptive sampling and selective data processing are used to reduce energy use while keeping good detection [7]. However, reducing monitoring can create risks. Lower detection sensitivity may lead to missed attacks, which can have serious consequences depending on the system.

The results further suggest that selective or adaptive monitoring may reduce energy use while maintaining high detection performance. However, actual carbon-aware scheduling was not implemented and remains future work.

The results show that network performance remained stable across all IDS models. Throughput changed only slightly, while delay increased during the attack phase. Since the throughput and delay values were identical across the three IDS models, the observed network performance changes are mainly caused by the attack traffic rather than by IDS model complexity. This means that the main trade-off in this simulation is between detection performance and energy use, not between IDS complexity and network quality.

A key challenge in evaluating sustainable IDS systems is that energy use and carbon

emissions are not always reported in a consistent way, and performance metrics do not always show how much energy a system uses [30]. This makes it difficult to compare systems and shows that both performance and energy should be measured together.

6.4 Unacceptable Security Risk

To evaluate whether carbon-aware scheduling or selective monitoring can be used without reducing security, it is necessary to clearly define what is meant by unacceptable security risk. According to the National Institute of Standards and Technology (NIST), security risk is defined as a function of the likelihood of an event and the impact of that event [39]. In this study, this relationship is operationalized as:

$$\text{Risk} = \text{Likelihood} \times \text{Impact} \quad (6.1)$$

where Risk represents the overall security risk, Likelihood represents the probability of occurrence of a threat event, and Impact represents the adverse consequences if the event occurs.

NIST does not define a fixed threshold for unacceptable risk. Instead, acceptable risk levels are determined by the organization based on its risk tolerance [39]. In intrusion detection research, system performance is usually evaluated using metrics such as detection rate, false alarm rate, and accuracy. These metrics show how well the system can separate normal traffic from malicious traffic [3]. Different detection methods achieve different results across these metrics, which means that improving one metric may affect another [3].

Anomaly-based intrusion detection systems can detect previously unseen attacks, but they require a balance between detection performance and resource usage, especially in carbon-aware settings [17]. In cyber-physical and energy systems, cyberattacks can disrupt normal operation and increase CO₂ emissions. A single attack scenario can result in emissions of several hundred tons of CO₂, while the emissions from continuous monitoring are much lower [40].

Based on these studies, this thesis defines unacceptable security risk as follows. Unacceptable security risk occurs when reduced monitoring leads to worse intrusion detection performance, for example lower detection accuracy or more missed attacks, to a level where the impact of undetected attacks becomes too high. In this definition, the impact includes both system effects and environmental effects.

6.5 Critical Discussion and Environmental Context

One challenge in this study was balancing simple models with realistic behavior. Simulations allow controlled and repeatable tests, but they cannot fully represent real networks and hardware. This affects the accuracy of both performance and energy results.

Another limitation is the use of simplified energy models. The energy values are based on estimates, not direct hardware measurements, which creates uncertainty in the exact numbers. However, comparisons between IDS models are still meaningful because the same method is used for all models.

The use of a centralized IDS makes analysis easier but limits how well the results apply to more complex or distributed systems. Also, the study focuses on one attack scenario,

which may not represent all real-world threats. These points should be considered when interpreting the results.

IDS systems can be considered part of always-on digital infrastructure, similar to data centres and cloud services. This is particularly relevant given global trends in electricity demand. Global electricity consumption increased by approximately 4.3% in 2024 (around 1100 TWh), with data centres and digital services contributing significantly to this growth [41]. This indicates that even relatively small systems such as IDS contribute to a rapidly expanding share of global energy use.

These observations should be interpreted alongside global emission trends. Energy-related CO₂ emissions reached approximately 37.8 Gt in 2024 [41], illustrating the continued coupling between energy use and environmental impact.

In contrast to well-documented emission sources such as transportation and national energy systems, the environmental impact of distributed digital infrastructure remains comparatively underreported. Public datasets on CO₂ emissions primarily focus on sectors such as transportation, industry, and national energy production [42, 43, 44], as well as vehicle-level emissions and fuel consumption [45, 46]. These datasets provide detailed and widely used benchmarks for environmental impact assessment.

However, comparable large-scale datasets for cybersecurity systems, including IDS deployments, are largely absent. This creates an asymmetry in how environmental impact is perceived: emissions from physical sectors are visible and routinely quantified, while emissions from distributed digital systems remain implicit and fragmented across millions of devices.

As a result, even though the per-system impact of IDS is small, their aggregate contribution is rarely considered in sustainability discussions. This suggests that cybersecurity infrastructure represents a form of “hidden” or “diffuse” energy consumption, where environmental impact is distributed across a large number of continuously operating systems rather than concentrated in a single sector.

Vehicle emission datasets indicate typical passenger car emissions of approximately 150 gCO₂/km [47, 45, 46]. This corresponds to approximately 35–40 km of driving per year, meaning that a single IDS system has a substantially lower environmental impact than even limited vehicle use.

However, the scaling effect is significant. At 10,000 deployments, IDS emissions reach approximately 55 tonnes CO₂ per year, while one million systems correspond to approximately 5,500 tonnes annually.

Table 6.1: Comparison between IDS emissions and equivalent passenger vehicle emissions

Source	CO ₂ emissions (kg/year)
Single IDS system	5.5
10,000 IDS systems	55,000
1 million IDS systems	5,500,000
Passenger vehicle (1,000 km/year)	150
Passenger vehicle (10,000 km/year)	1,500
Passenger vehicle (15,000 km/year)	2,250

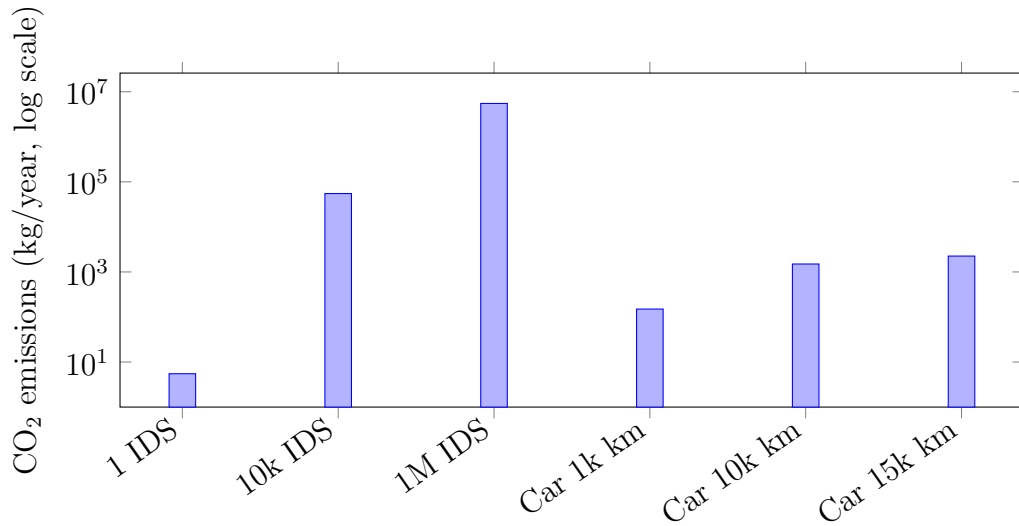


Figure 6.1: Comparison between IDS emissions and passenger vehicle emissions (log scale).

The comparison highlights the relative scale of IDS-related emissions in relation to everyday activities. A single IDS system produces approximately 5.5 kg CO₂ per year, which is significantly lower than even limited passenger vehicle use. However, the scaling effect is substantial: 10,000 systems correspond to emissions comparable to tens of passenger vehicles, while one million systems correspond to emissions on the order of several thousand vehicles.

7 | Conclusion

This thesis studied the trade-offs between detection performance, energy use, and environmental impact in intrusion detection systems used in Wi-Fi networks. The study compared Static, Adaptive, and Hybrid IDS models in controlled simulations to understand how IDS design affects both security and sustainability.

The results show that IDS design clearly affects both detection performance and energy consumption. The Hybrid IDS achieved the best detection performance, with an F1-score of 0.9565, but it also consumed the most energy (8.21×10^{-7} kWh) and produced the highest carbon emissions. The Static IDS consumed the least energy (8.12×10^{-7} kWh), but its low precision (0.5789) resulted in weaker overall detection performance, with an F1-score of 0.7333. The Adaptive IDS provided the most balanced result, achieving high detection performance with an F1-score of 0.9524 and moderate energy consumption (8.16×10^{-7} kWh), lower than the Hybrid IDS.

One important finding is that increasing IDS complexity gives diminishing performance benefits. The improvement from the Adaptive IDS to the Hybrid IDS is small, while energy use still increases by approximately 0.51%. This indicates that detection performance does not increase at the same rate as model complexity, while energy consumption increases more directly. This is due to increased computational and communication overhead, where additional processing leads to higher energy consumption. The efficiency results support this interpretation, since the Adaptive and Hybrid IDS have almost the same efficiency (1.17×10^6 F1-score per kWh), while the Static IDS is much less efficient.

From a system perspective, network performance is not strongly affected by IDS complexity. Throughput decreases only slightly during attacks, by approximately 0.42%, while delay increases substantially, by approximately $13.8\times$. This increase is mainly caused by attack traffic, not by the IDS itself. This means that the main trade-off observed in this study is between detection performance and energy use, not between IDS complexity and network quality.

The main contribution of this thesis is a quantitative analysis of IDS performance, energy use, and carbon emissions within one experimental setup. The results show that medium-complexity IDS designs, such as the Adaptive IDS, can achieve near-optimal detection performance without unnecessary energy use. This demonstrates that energy-efficient IDS design is possible without significant loss in detection performance.

The research questions are answered as follows. First, IDS design has a strong effect on energy use and carbon emissions, because more complex models require more computation. This is due to increased computational and communication overhead, which directly increases energy consumption. Second, in the centralized system used in this study, system behavior

depends more on model complexity than on system architecture. However, this result is limited to the evaluated centralized setup, and the impact of other architectures remains an open question. Third, adaptive monitoring can reduce energy use while keeping high detection performance, which shows that carbon-aware IDS systems are possible without large security risks.

At a larger scale, the results show that size matters. A single IDS uses little energy, but continuous operation and large deployments can lead to noticeable environmental impact. This means that IDS systems are part of hidden energy use in digital infrastructure.

The results are supported by consistent behavior across multiple simulation runs and low variation in measurements. However, there are limitations. The study uses simulations, simplified network conditions, and estimated energy and carbon values. These factors affect the exact values, but the differences between IDS models are still reliable because the experiments are consistent.

In conclusion, strong intrusion detection does not require very complex models. Instead, adaptive IDS designs can achieve near-optimal detection performance while avoiding unnecessary increases in energy consumption. Well-designed adaptive IDS systems can provide near-best performance with lower energy use. Future IDS design should consider multiple goals, including security, efficiency, and environmental impact.

7.1 Generalization of the Results

The results are mainly valid for simulated Wi-Fi networks with centralized access-point-level IDS. However, the main findings can be applied to other systems where IDS runs continuously and uses computational resources.

The trade-off between detection performance and energy use is likely to exist in many IDS and cybersecurity systems. The idea that increasing complexity gives smaller benefits is also common in machine learning and system design.

Even if the exact numbers depend on the situation, the main idea remains: medium-complexity models can achieve near-best performance with lower resource use. This makes the results useful for designing energy-efficient and sustainable cybersecurity systems in general.

The absolute energy and carbon values may vary in real deployments depending on hardware, traffic load, wireless conditions, IDS implementation, and regional electricity mix. However, the relative pattern observed in this study is relevant: increased IDS complexity tends to increase energy consumption, while adaptive monitoring can provide a more balanced trade-off between detection performance and sustainability.

7.2 Limitations and Future Work

The results are based on a controlled simulation environment and should therefore be interpreted as comparative rather than directly representative of all real-world Wi-Fi deployments. Simulations allow controlled and repeatable tests, but they cannot fully represent real networks and hardware. This affects the accuracy of both performance and energy results.

Another limitation is the use of simplified energy models. The energy values are based on estimates, not direct hardware measurements, which creates uncertainty in the exact

numbers. However, comparisons between IDS models are still meaningful because the same method is used for all models.

The use of a centralized IDS makes analysis easier but limits how well the results apply to more complex or distributed systems. Also, the study focuses on one attack scenario, which may not represent all real-world threats. These points should be considered when interpreting the results.

Future work should study other IDS architectures, including distributed and hybrid systems, to examine how system design affects energy use and detection performance. In particular, future studies should compare AP-level, edge/controller-level, and centralized/cloud-based IDS deployments to provide a more complete answer to how monitoring architecture affects both security performance and sustainability.

Real-world experiments are also needed to confirm the simulation results and improve the accuracy of energy models. Hardware-based measurements would make it possible to validate the estimated energy consumption and carbon emission values.

Another important direction is to treat IDS design as a multi-objective optimization problem, where detection performance, delay, and energy use are optimized together. This could help create adaptive IDS systems that adjust their behavior based on system conditions and environmental factors, such as real-time carbon intensity.

Future work should also implement actual carbon-aware scheduling rather than only estimating carbon impact after simulation. This could include dynamically adjusting monitoring intensity based on regional or temporal carbon intensity, workload, and security risk.

Future research should also explore more advanced adaptive methods, such as machine learning-based detection, dynamic thresholds, and selective monitoring. Large-scale studies are also needed to better understand the total environmental impact of IDS systems in realistic deployment scenarios.

Finally, future research should investigate large-scale datasets for cybersecurity energy use and carbon emissions. Such datasets would make it easier to compare cybersecurity systems with other sectors and would improve transparency around the environmental impact of distributed digital security infrastructure.

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Glossary and Symbols

Carbon Footprint (C) Total greenhouse gas emissions associated with system operation, computed from energy consumption and carbon intensity.

Carbon Intensity (κ) Amount of CO₂ emissions per unit of electrical energy (e.g., gCO₂/kWh), dependent on the regional energy mix.

Sustainability (S) Aggregate measure of environmental impact, represented in this study by energy consumption and carbon emissions.

Energy Consumption (E) Total electrical energy used during a simulation run, measured at the access point.

Computational Load (L) Processing demand introduced by IDS operations, contributing to overall energy consumption.

Throughput (τ) Rate of successful data delivery over the network, measured in bits per second.

Detection Rate / Recall (R) Proportion of actual attacks correctly identified:

$$R = \frac{TP}{TP + FN}$$

Precision (P) Proportion of detected attacks that are correct:

$$P = \frac{TP}{TP + FP}$$

F1-score (F_1) Harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

Deep Packet Inspection (DPI) Packet-level analysis method examining payload content; limited in effectiveness for encrypted traffic.

Wireless Intrusion Detection System (WIDS) System designed to monitor and detect anomalies in wireless network traffic.

Flow-Based Monitoring Monitoring approach based on aggregated traffic flows rather than individual packets, enabling scalable analysis.

Packet Capture Packet-level monitoring method providing detailed traffic data at the cost of higher resource usage.